

# VARIANCE-OPTIMAL CONTROL VARIATES FOR BLACK-BOX LEARNING

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## ABSTRACT

More and more system learn through components that are cannot differentiate, such as a black-box solver. In this case, gradients are often estimated using score-function (REINFORCE) estimator. While unbiased, these estimators can have prohibitively high variance. A standard variance-reduction strategy is to learn a decision-specific value function and use it as a control variate. In this paper, we first observe that even if exact value function can be far from variance-optimal. Its resulting gradient variance can be arbitrarily larger than that of the optimal control variate. We show that this gap arises because conventional value-function control variates treat decisions separately, while their gradient contributions interact through shared parameters. Motivated by this observation, we derive a closed-form correction that transforms the value-function control variate into the variance-optimal one. We further characterize when this correction provides substantial variance reduction.

## 1 INTRODUCTION

Many learning systems need to learn through a component that is not differentiable. A predictor reads a context and outputs a vector of parameters, and then a black-box oracle evaluates the parameters and returns a scalar loss. In decision-focused learning, the predicted parameters define an optimization problem, and the loss is the regret of the decision made by the solver. In LLM agents, the predicted parameters can be a tool configuration, e.g., which Model Context Protocol (MCP) servers the agent allows to call. In all these cases, the loss is opaque, so we have no access to its gradient, and each oracle call can potentially be expensive.

A standard solution is the score function (REINFORCE) estimator, which requires only one oracle evaluation and gives an unbiased gradient. Directly applying it often leads to two major challenges: 1) high variance, which grows with the number of predicted parameters, and 2) equal credit for each parameter, since all of them receive the same scalar loss. Recently, exact local marginalization methods are proposed to reduce the variance of REINFORCE by removing the sampling noise of one parameter at a time by evaluating the loss under every possible value of that parameter (). This provides better credit assignment, but requires up to  $dV$  oracle evaluations for  $d$  parameters with  $V$  possible values, which is impractical when the *oracle is expensive*.

A seemingly natural alternative is to learn a differentiable surrogate of the black-box loss and replace the oracle with it (). This amortizes the cost of enormous numbers of oracle calls, but naively differentiating through the learned surrogate generally produces a biased gradient, as the approximation error in the surrogate directly changes the optimization objective, leading to potential convergence inconsistency. The resulting trade-offs are summarized in Table 1.

These trade-offs motivate this paper: *How can we efficiently learn through a black-box oracle with expensive evaluations when it returns only a single loss for many decisions?* Common ways to address this credit-assignment problem is to learn a decision-specific value function (). Instead of using the same observed loss for each decision, the value function predicts the expected loss associated with each possible choice. When used as an action-dependent control variate, this prediction can provide fine-grained credit while preserving an unbiased gradient.

However, we first challenge this by observing that it can have a large variance when the weights are shared, such as one network predicts all parameters of the downstream problem, or auto-regressively

Table 1: Trade-offs for optimizing discrete predictions through an expensive black-box oracle.

|                        | REINFORCE | Exact local marg. | Learned surrogate | Ours         |
|------------------------|-----------|-------------------|-------------------|--------------|
| Black-box oracle calls | 1         | $dV$              | 1                 | 1            |
| Unbiased gradient      | ✓         | ✓                 | ×                 | ✓            |
| Variance of estimator  | High      | Low               | Low               | Low          |
| Credit assignment      | Poor      | Fine-grained      | Fine-grained      | Fine-grained |

generated parameters. With shared parameters, the gradient contributions from different decisions interact in the same parameter space. As a result, minimizing the variance of each decision separately does not necessarily minimize the variance of their sum. We show that even a perfectly learned conditional value function can therefore have arbitrarily higher variance than the globally optimal control variate.

In addition, we derive a closed-form for optimal control variate. The key insight is that the conditional value function solves only the local credit-assignment problem, which only captures how the expected loss changes with each individual decisions. However, the variance of the fully gradient also depends on how the gradient contributions of different decisions interact. The optimal control variate therefore need to consist of the usual conditional value function together with an **additional correction** that accounts for this cross-decision effects. When decisions are effectively decoupled, the correction vanishes and the standard value function is optimal. When their gradient contributions are strongly coupled, the correction can substantially reduce the remaining variance.

## 2 RELATED WORKS

**Black-box gradient estimation.** When the loss is not differentiable, the score-function estimator (Williams, 1992) is the standard unbiased choice, and Silvestri et al. (2026) recently apply it to DFL. Due to the high variance nature of the score function, a line of work has been proposed to reduce the variance by baselines, including the optimal constant baselines (), learned baselines (), and leave-one-out baselines (). Group-relative baselines are now widely used for LLM reinforcement learning (), and Li et al. (2026) derive the optimal token-level baseline for long-horizon LLM RL. (Cheng et al., 2020)

However, these baselines do not depend on action, so every coordinate still receives the same credit. Action-dependent control variate removes this limitation, and they have been build from continuous relaxation (), Stein’s identity (), factorized policies (), and discrete structures (), although show their gain over an action-independent baseline is small in practice. These action-dependent baselines mainly propose new forms of control variate, but

**Exact marginalization.** Another line of work sums out one variable at a time instead of sampling it (). In our

(Liu et al., 2019; Smet et al., 2023; ?)

A parallel line is also from black-box for stochastic optimization in variational inference, where the log derivative trick is also employed to estimate the ELBO (AUEB et al., 2015)

## 3 PROBLEM STATEMENT

**Black-box learning setting.** For a context  $x$ , a stochastic predictor parameterized by  $\theta$  samples a (discrete) parameter vector  $b = (b_1, \dots, b_d)$ , according to a known categorical distribution

$$p(b|x, \theta) = \prod_{j=1}^d p(b_j|z_j, x, \theta), \quad (1)$$

where  $z_j$  denotes the context available when decision  $j$  is sampled. We consider two common cases in practice. For a *factorized* predictor, the coordinates are independent given  $x$ , and we set  $z_j = b_{-j}$

when analyzing coordinate  $j$ . Note that  $p_\theta(b_j|z_j, x) = p_\theta(b_j|x)$  in this case. This is the standard predictor in DFL, which outputs all parameters at once. For an *autoregressive* predictor such as an LLM that outputs one token after another, we set  $z_j = b_{<j}$ . In both cases, we only use the fact that  $b_j$  is sampled from a known categorical distribution given  $z_j$ .

The sampled vector is then evaluated by a black-box oracle, yielding the following problem:

$$\min_{\theta} \mathcal{L}(\theta) := \mathbb{E}_{p(b|\theta)}[f(b)] = \mathbb{E}_{p(b|\theta)}[\text{Eval}(\text{Oracle}(b))]. \quad (2)$$

In DFL, Oracle solves a downstream problem with the predicted parameter  $b$ , and Eval evaluates the decision under the true parameters. Our analysis only requires  $f$  can be evaluated but not differentiated.

**REINFORCE.** Since the Oracle is a black-box, we use the score-function identity

$$g := \nabla_{\theta} \mathcal{L}(\theta) = \mathbb{E}[f(b) \sum_{j=1}^d s_j(b_j)], \quad s_j(v) := \nabla_{\theta} \log p_j(v). \quad (3)$$

The single-sample REINFORCE estimator requires only one evaluation of  $f$ , but it has two familiar limitations: high variance and coarse credit assignment, since the same sampled loss multiplies every score term.

**Scope of the analysis.** The one-coordinate marginalization results in the next section apply to both factorized and autoregressive predictors. For the general residual-variance decomposition, we specialize to the autoregressive case  $z_j = b_{<j}$ . The nested contexts allow the remaining gradient noise to be decomposed decision by decision without introducing additional assumptions on the black-box loss.

**Notation.** We use the shorthand  $p_j(v) = p_\theta(b_j = v | z_j)$  and  $s_j(v) = \nabla_{\theta} \log p_j(v)$ .

## 4 VARIANCE-OPTIMAL CONTROL VARIATE

We ask how much of the score-function variance can be removed without introducing bias. Our key observation is that the coordinate-wise optimality can be strictly suboptimal when parameters are shared across decisions. Then, we characterize the remaining variance through prefix-wise residuals and derive a variance-optimal correction in closed form.]

### 4.1 FROM ONE-COORDINATE MARGINALIZATION TO AN UNBIASED ESTIMATOR

Consider the  $j$ -th term in Equation (3). Let  $b_{\text{rem}(j)}$  denote the randomness not contained in  $(z_j, b_j)$ . Applying the law of total expectation gives

$$\begin{aligned} \mathbb{E}_b[f(b)s_j(b_j)] &= \mathbb{E}_{z_j} \mathbb{E}_{b_j|z_j} \mathbb{E}_{b_{\text{rem}(j)}|z_j, b_j} [f(b)s_j(b_j)] \\ &= \mathbb{E}_{z_j} \mathbb{E}_{b_j|z_j} \left[ \underbrace{\mathbb{E}[f(b)|z_j, b_j]}_{Q_j(z_j, b_j)} s_j(b_j) \right], \end{aligned} \quad (4)$$

where

$$Q_j(z_j, v) := \mathbb{E}[f(b) | z_j, b_j = v]. \quad (5)$$

Since  $b_j$  is categorical,

$$g = \sum_{j=1}^d \mathbb{E}_{z_j} \left[ \sum_{v \in \mathcal{V}} p_j(v) Q_j(z_j, v) s_j(v) \right]. \quad (6)$$

Here,  $Q_j$  is an one-coordinate conditional mean, which is widely used in RL for variance reduction (). This is the same basic marginalization principle used by categorical gradient estimators, which sum explicitly over its possible values and average over the remaining randomness.

The exact sweep is usually too expensive to compute, so a common way is to replace  $Q_j$  by fitting a predictor  $\hat{Q}_j(z_j, v) \approx Q_j(z_j, v)$  from previous oracle evaluations.

However, simply replacing  $Q_j$  by  $\hat{Q}_j$  in Equation (6) would introduce bias due to approximation error. Instead, we use  $\hat{Q}_j$  as a control variate. Adding and subtracting the expectation gives

$$\mathbb{E}_b[f(b)s_j] = \mathbb{E}_b[(f(b) - \hat{Q}_j(z_j, b_j))s_j(b_j)] + \mathbb{E}_{z_j} \left[ \sum_{v \in \mathcal{V}} p_j(v) \hat{Q}_j(z_j, v) s_j(v) \right].$$

The second term contains no black-box evaluation, because for a sampled context  $z_j$ , we can enumerate every  $V$  possible values of  $b_j$  and compute this expectation exactly. Therefore, a single sampled trajectory  $b$  gives the following estimator:

$$\hat{g}_{\hat{Q}} := \sum_{j=1}^d \left[ (f(b) - \hat{Q}(z_j, b_j)) s_j + \sum_{v \in \mathcal{V}} p_j(v) \hat{Q}(z_j, v) s_j(v) \right]. \quad (7)$$

**Lemma 4.1** (Unbiasedness and coordinate-wise optimality). *Given a fitted predictor  $\hat{Q}$ , Equation (7) is unbiased for any choice of  $\hat{Q}$ :*

$$\mathbb{E}[\hat{g}_c] = \nabla_{\theta} \mathcal{L}(\theta) = g. \quad (8)$$

Moreover, for a fixed decision  $j$  and context  $z_j$ , the exact conditional mean  $Q_j$  minimizes the conditional variance of the  $j$ -th summand in Equation (7) over all functions of  $(z_j, b_j)$ .

*Proof.* For unbiasedness, conditioning on  $z_j$  gives

$$\mathbb{E}_{b_j|z_j}[\hat{Q}_j(z_j, b_j) s_j(b_j)] = \sum_{v \in \mathcal{V}} p_j(v) \hat{Q}_j(z_j, v) s_j(v), \quad (9)$$

so the two terms involving  $\hat{Q}_j$  cancel in expectation.

For the second claim, consider any perturbation  $Q_j + \delta_j$ . Let  $\hat{g}_j[Q]$  denote the  $j$ -th summand obtained with the exact  $Q_j$ , and define

$$u_j := \delta_j(z_j, b_j) s_j(b_j) - \sum_v p_j(v) \delta_j(z_j, v) s_j(v). \quad (10)$$

Then

$$\hat{g}_j[Q + \delta] = \hat{g}_j[Q] - u_j, \quad \mathbb{E}[u_j | z_j] = 0. \quad (11)$$

At the exact conditional mean,

$$\hat{g}_j[Q] - \mathbb{E}[\hat{g}_j[Q] | z_j] = (f(b) - Q_j(z_j, b_j)) s_j(b_j). \quad (12)$$

Since  $\mathbb{E}[f - Q_j | z_j, b_j] = 0$ , this residual is conditionally orthogonal to  $u_j$ . Hence

$$\text{Var}(\hat{g}_j[Q + \delta] | z_j) = \text{Var}(\hat{g}_j[Q] | z_j) + \mathbb{E}[\|u_j\|_2^2 | z_j], \quad (13)$$

which is minimized by the exact conditional mean (up to perturbations that induce the same zero correction).  $\square$

The lemma explains why  $Q_j$  is the natural target for a learned action-dependent control variate. It removes all variance that can be removed when decision  $j$  is optimized in isolation. However, we do not know whether  $Q_j$  is variance-optimal.

## 4.2 WHY THE CONDITIONAL MEAN IS NOT GLOBAL OPTIMALITY

The conditional mean  $Q$  is the natural first choice, but the previous derivation only says that it removes two loss for each coordinate. It does *not* yet say that the sum of all coordinate gradient terms has minimum variance. **Zihao: The example may need to be the arbitrary worse one + autoregressive (can be in Appendix).**

**Example 4.2.** Consider two independent Bernoulli variables  $b_1, b_2 \sim \text{Bernoulli}(p)$  with a shared parameter  $p = \sigma(\theta)$ . Let  $f(b_1, b_2) = b_1 b_2$ . At  $\theta = 0$ , we have  $p = 1/2$  and  $\nabla_\theta p = 1/4$ , so

$$\nabla_\theta \mathcal{L}(\theta) = \nabla_\theta \mathbb{E}_{p(\theta)}[f] = \frac{1}{4}. \quad (14)$$

For the condition-mean control variates,

$$Q_1(b_2, v) = v b_2 \quad Q_2(b_1, v) = b_1 v, \quad (15)$$

and hence  $Q_j(b_{-j}, b_j) = f(b)$ .

Intuitively,  $Q_1$  averages over all possible values of  $b_1$ , so the randomness from  $b_1$  is removed; however,  $b_2$  is still sampled and remains random. Similarly,  $Q_2$  removes the randomness from  $b_2$  but not from  $b_1$ . Thus, substituting  $c = Q$  gives

$$\hat{g}[Q] = \frac{1}{4} b_2 + \frac{1}{4} b_1 = \frac{1}{4} (b_1 + b_2), \quad (16)$$

and thus

$$\text{Var}(\hat{g}[Q]) = \frac{1}{32} > 0. \quad (17)$$

Now consider  $c_1 = Q_1 + 1/4$ ,  $c_2 = Q_2 + 1/4$ . Since  $\nabla_\theta \log p(b_j) = b_j - \frac{1}{2}$ , the additional terms exactly cancel the remaining randomness:

$$\hat{g}[c] = \frac{1}{4} (b_1 + b_2) - \frac{1}{4} \left[ \left( b_1 - \frac{1}{2} \right) + \left( b_2 - \frac{1}{2} \right) \right] = \frac{1}{4}. \quad (18)$$

Hence,

$$\text{Var} \hat{g}[c] = 0 < \text{Var}(\hat{g}[Q]). \quad (19)$$

The issue in Theorem 4.1 is not an inaccurate value function, because even  $Q$  is exact, its variance could still be suboptimal. Rather,  $Q_j$  minimizes the variance of one coordinate at a time, while the quantity we ultimately care about is the variance of their sum in a shared parameter space. We next characterize exactly what remains.

#### 4.3 A GENERAL FRAMEWORK FOR VARIANCE-OPTIMAL CONTROL VARIATES

Let  $e_Q := \hat{g}_Q - g$  denote the remaining noise after subtracting  $Q$ . We first decompose the residual  $e_Q$  into different decision time step:

$$\xi_j(z_j, v) := \mathbb{E}[\hat{g}_Q | z_j, b_j = v] - \mathbb{E}[\hat{g}_Q | z_j]. \quad (20)$$

Thus,  $\xi_j(z_j, v)$  measures how much choosing  $b_j = v$  can change the expected gradient. By the law of total probability, we have

$$\mathbb{E}_{b_j | z_j}[\xi_j(z_j, v)] = \sum_v p_j(v) \xi_j(z_j, v) = 0. \quad (21)$$

**Lemma 4.3** (Decision-wise decomposition). *Along a sampled trajectory  $b$ , we can write*

$$e_Q = \hat{g}_Q - g = \sum_{j=1}^d \xi_j(z_j, b_j). \quad (22)$$

Moreover, for any  $i < j$ , we have

$$\mathbb{E}[\xi_i(z_i, b_i)^\top \xi_j(z_j, b_j)] = 0. \quad (23)$$

Therefore, the variance of  $\hat{g}_Q$  is

$$\text{Var}(\hat{g}_Q) = \sum_{j=1}^d \mathbb{E}[\|\xi_j(z_j, b_j)\|_2^2]. \quad (24)$$

*Proof.* Because  $z_j = b_{<j}$  and  $(z_j, b_j) = b_{\leq j}$ ,

$$\begin{aligned} \sum_{j=1}^d \xi_j(z_j, b_j) &= \sum_{j=1}^d (\mathbb{E}[\hat{g}_Q \mid b_{\leq j}] - \mathbb{E}[\hat{g}_Q \mid b_{<j}]) \\ &= \hat{g}_Q - \mathbb{E}[\hat{g}_Q] = e_Q, \end{aligned} \quad (25)$$

where the sum telescopes. For any  $i < j$ ,  $\xi_i(z_i, b_i)$  is determined by  $b_{\leq i}$  and hence by  $z_j = b_{<j}$ . Therefore,

$$\mathbb{E}[\xi_i^\top \xi_j] = \mathbb{E}[\xi_i^\top \mathbb{E}[\xi_j \mid z_j]] = 0, \quad (26)$$

where the last equality follows from Equation (21). Equation (24) then follows by expanding the squared norm of Equation (22).  $\square$

**Where does the remaining variance come from?** Lemma 4.3 shows that the residual variance is fully captured by  $\xi_j$ . For an autoregressive model, revealing  $b_j$  can affect gradient terms both before and after  $j$ . Specifically,

$$\xi_j(z_j, v) = \left( Q_j(z_j, v) - \sum_w p_j(w) Q_j(z_j, w) \right) \sum_{i < j} s_i(b_i) + \mathbb{E}[\hat{g}_{Q, > j} \mid z_j, b_j = v] - \mathbb{E}[\hat{g}_{Q, > j} \mid z_j], \quad (27)$$

where  $\hat{g}_{Q, > j} := \sum_{i > j} \hat{g}_{Q, i}$ . The first term captures the effect of  $b_j$  on earlier score terms, while the second captures its effect on future gradient terms. These are the residual cross-decision effects left by  $Q$ .

**General control variate.** We can now turn this decomposition into a control-variate design principle. Let  $\Delta_j(z_j, v)$  be any gradient-shaped function. We center it over the current decision to obtain a zero-mean function:

$$\tilde{\Delta}_j := \Delta_j(z_j, v) - \sum_v p_j(v) \Delta_j(z_j, v), \quad U_\Delta := \sum_{j=1}^d \tilde{\Delta}_j. \quad (28)$$

By construction,  $\mathbb{E}[\tilde{\Delta}_j \mid z_j] = 0$  and hence  $\mathbb{E}[U_\Delta] = 0$ . Therefore,

$$\hat{g}_\Delta := \hat{g}_Q - U_\Delta \quad (29)$$

is unbiased for any choice of  $\Delta$ .

**Theorem 4.4.** For any collection of gradient-valued control variates  $\Delta = (\Delta_1, \dots, \Delta_d)$ ,

$$\text{Var}(\hat{g}_\Delta) = \text{Var}(Q) + \sum_{j=1}^d \mathbb{E}[\|\tilde{\Delta}_j\|^2 - 2\xi_j^\top \tilde{\Delta}_j] = \sum_{j=1}^d \mathbb{E}\|\xi_j(z_j, b_j) - \tilde{\Delta}_j(z_j, b_j)\|^2. \quad (30)$$

Hence, among unrestricted control variates, the optimum one is

$$\tilde{\Delta}_j^*(z_j, v) = \xi_j(z_j, v), \quad j = 1, \dots, d. \quad (31)$$

*Proof.* See Section A.1 for proof. The key idea is  $\square$

Theorem 4.4 shows that, after applying the conditional-mean control variate  $Q$ , the remaining variance is exactly the mismatch between the residual increment  $\xi_j$  and the additional control variate. Therefore, the ideal correction is simply  $\tilde{\Delta}_j = \xi_j$ . In practice,  $\xi_j$  is not directly realizable by the model gradient, so we next restrict the control variate to the row-wise gradient subspace.

Thus, corrections constructed at different prefixes do not interfere with each other. We can design them one prefix at a time and then add them together.

#### 4.4 MAIN RESULT: THE GLOBAL VARIANCE OPTIMUM

Theorem 4.4 characterizes the ideal control variate in gradient space, but does not yet specify a practical representation. Neural-network gradients provide additional structure. Consider a linear weight matrix and one of its parameter rows  $r$ . At decision  $j$ , the gradient within that row lies along the corresponding layer input. Let  $x_{j,r}$  denote this local input direction, which is fixed by the prefix  $z_j$  and therefore does not depend on the candidate action  $v$ . The realizable row-wise subspace is

$$\mathcal{S}_{j,r} := \text{span}\{x_{j,r}\}. \quad (32)$$

We restrict the centered control variate for row  $r$  to this subspace. Writing  $\xi_{j,r}$  for the component of  $\xi_j$  corresponding to row  $r$ , Equation (30) shows that the optimal row-wise control variate is the point in  $\mathcal{S}_{j,r}$  closest to  $\xi_{j,r}$ .

**Theorem 4.5.** Let  $\Pi_{j,r} := \frac{x_{j,r}x_{j,r}^\top}{\|x_{j,r}\|_2^2}$ , with  $\Pi_{j,r} = 0$  when  $x_{j,r} = 0$ . Then the variance-minimizing row-wise control variate is

$$\Delta_{j,r}^*(v) = \Pi_{j,r}\xi_j(v). \quad (33)$$

It is automatically centered over the current decision, and the resulting estimator satisfies

$$\text{Var}(\hat{g}_{\text{row}}) = V_Q - \sum_{j=1}^d \mathbb{E} \left[ \sum_r \|\Pi_{j,r}\xi_{j,r}(z_j, b_j)\|_2^2 \right]. \quad (34)$$

*Proof.* For a fixed  $(j, r)$ , Equation (30) reduces the optimization to

$$\arg \min_{u \in \mathcal{S}_{j,r}} \|u - \xi_{j,r}(z_j, v)\|_2^2, \quad (35)$$

whose solution is the Euclidean projection in ???. Since  $\Pi_{j,r}$  depends on the prefix and row but not on  $v$ ,

$$\sum_v p_j(v) \tilde{\rho}_{j,r}^*(z_j, v) = \Pi_{j,r} \sum_v p_j(v) \xi_{j,r}(z_j, v) = 0, \quad (36)$$

where the last equality follows from Equation (21). Thus the projected control variate is already centered. Finally, the Pythagorean identity for orthogonal projection gives

$$\|\xi_{j,r} - \Pi_{j,r}\xi_{j,r}\|_2^2 = \|\xi_{j,r}\|_2^2 - \|\Pi_{j,r}\xi_{j,r}\|_2^2. \quad (37)$$

Summing over rows and decisions and using Equation (24) yields ??.

## 5 EXPERIMENTS

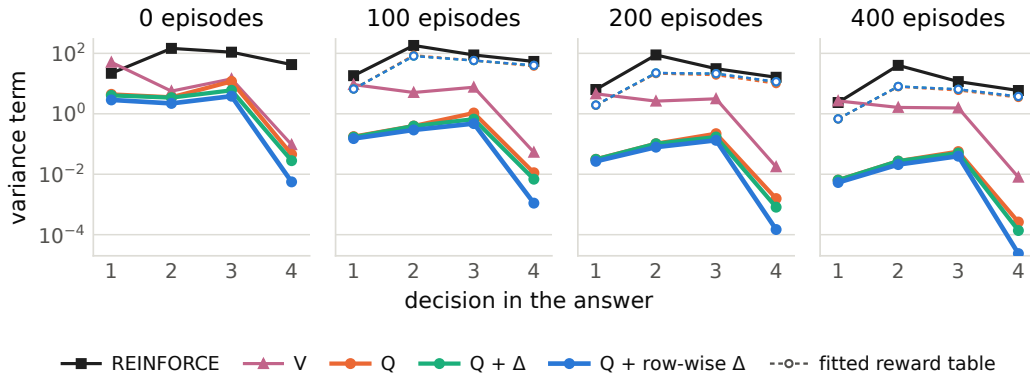


Figure 1: Caption

Ablation study:

Left: 1. The test before training: cos sim between score functions. 2. The cost of the reward model. Vary the offline dataset size, fit  $\hat{f}$ , and plot the variance gap to the exact table.

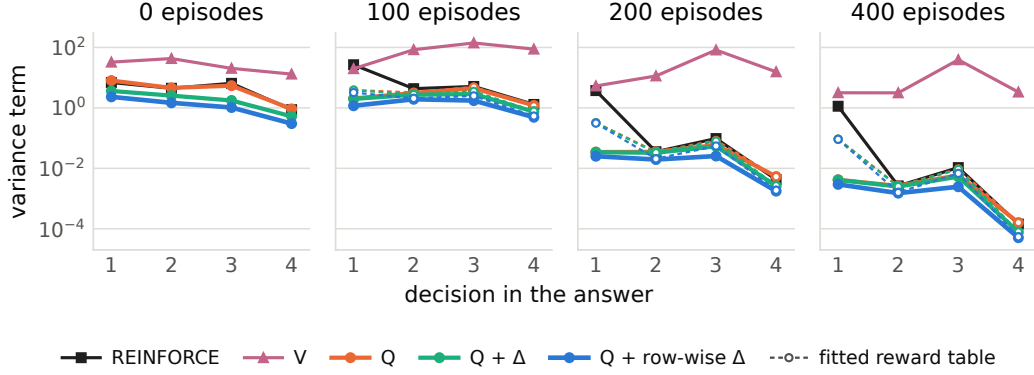


Figure 2: Caption

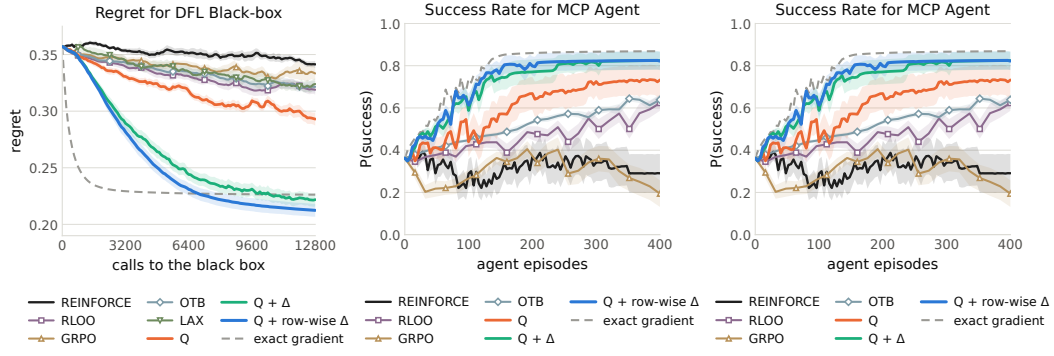


Figure 3: Regret performance

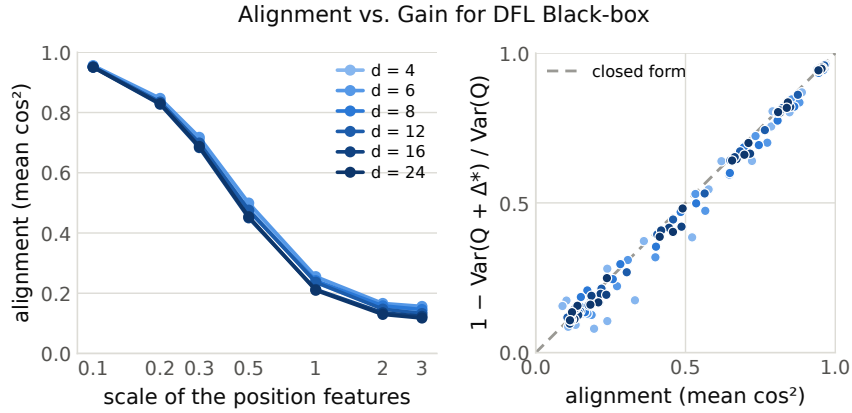


Figure 4: Caption

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## ACKNOWLEDGMENTS

Use unnumbered third level headings for the acknowledgments. All acknowledgments, including those to funding agencies, go at the end of the paper.

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## A APPENDIX

You may include other additional sections here.

We can then decompose  $\hat{g}_Q$  by different time step. For  $i < j$ ,  $s_i(b_i)$ ,  $Q_i(z_i, b_i)$  are already determined by  $b_{\leq j}$ . The only random object is  $f(b)$ , and by the definition of  $Q_j$  in Equation (5), we have

$$Q_j(v) \underbrace{\sum_{i < j} s_i(b_i)}_{S_{< j}} - \underbrace{\sum_{i < j} Q_i(z_i, b_i) s_i(b_i) + \sum_{i < j} \sum_v p_i(w) Q_i(z_i, v) s_i(v)}_{:= R_j \text{ (does not depend on } v)}. \quad (38)$$

Thus, we can rewrite Equation (7) as

$$\mathbb{E}[\hat{g}_Q | z_j, b_j = v] = R_j + Q_j S_{< j} + D_j, \quad (39)$$

where  $D_j(v) = \mathbb{E}[\sum_{i > j} ((f(b) - Q_i(z_i, b_i)) s_i(b_i) + \sum_w p_i(w) Q_i(z_i, w) s_i(w)) | z_j, b_j = v]$  collects the remaining conditional expected gradient after choosing  $v$ . Similarly, we have

$$\mathbb{E}[\hat{g}_Q | z_j] = R_j + V_j S_{< j} + \bar{D}_j, \quad (40)$$

where  $\bar{D}_j = \sum_v p_j(v) D_j(v)$ . Thus, we have

$$\xi_j(z_j, v) := \mathbb{E}[\hat{g}_Q | z_j, b_j = v] - \mathbb{E}[\hat{g}_Q | z_j] = (Q_j - V_j) S_{< j} + D_j - \bar{D}_j. \quad (41)$$

**Remark A.1** (Relation to policy gradients). Our gradient estimator looks very alike to the standard policy gradient. But we consider 1) marginalization the randomness in each coordinator  $j$ ; 2) standard policy gradient doesn't have extra terms because they only consider the expectation of the gradient estimator instead of variance minimization.

### A.1 PROOF FOR THEOREM 4.4

*Proof.* Let

$$e_Q := \hat{g}_Q - g. \quad (42)$$

Since  $\hat{g}_\Delta = \hat{g}_Q - \sum_j \tilde{\Delta}_j$  is unbiased,

$$\begin{aligned} \text{Var}(\hat{g}_\Delta) &= \mathbb{E}[\|e_Q - \sum_{j=1}^d \tilde{\Delta}_j\|_2^2] \\ &= V_Q - 2 \sum_{j=1}^d \mathbb{E}[e_Q^\top \tilde{\Delta}_j] + \mathbb{E}[\|\sum_{j=1}^d \tilde{\Delta}_j\|_2^2]. \end{aligned} \quad (43)$$

For  $i < j$ ,  $\tilde{\Delta}_i$  is determined by  $z_j = b_{<j}$ , while  $\mathbb{E}[\tilde{\Delta}_j \mid z_j] = 0$ . Hence

$$\mathbb{E}[\tilde{\Delta}_i^\top \tilde{\Delta}_j] = \mathbb{E}[\tilde{\Delta}_i^\top \mathbb{E}[\tilde{\Delta}_j \mid z_j]] = 0, \quad (44)$$

so

$$\mathbb{E}[\|\sum_{j=1}^d \tilde{\Delta}_j\|_2^2] = \sum_{j=1}^d \mathbb{E}[\|\tilde{\Delta}_j\|_2^2]. \quad (45)$$

Next, since  $\tilde{\Delta}_j$  depends only on  $(z_j, b_j)$ ,

$$\mathbb{E}[e_Q^\top \tilde{\Delta}_j] = \mathbb{E}[\mathbb{E}[e_Q \mid z_j, b_j]^\top \tilde{\Delta}_j]. \quad (46)$$

By the definition of  $\xi_j$ ,

$$\mathbb{E}[e_Q \mid z_j, b_j] = \xi_j(z_j, b_j) + \mathbb{E}[\hat{g}_Q \mid z_j] - g. \quad (47)$$

The last two terms depend only on  $z_j$ , and therefore vanish after taking the inner product with  $\tilde{\Delta}_j$ . Thus,

$$\mathbb{E}[e_Q^\top \tilde{\Delta}_j] = \mathbb{E}[\xi_j^\top \tilde{\Delta}_j]. \quad (48)$$

Substituting Equations (45) and (48) into Equation (43) gives

$$\text{Var}(\hat{g}_\Delta) = V_Q + \sum_{j=1}^d \mathbb{E}[\|\tilde{\Delta}_j\|_2^2 - 2\xi_j^\top \tilde{\Delta}_j]. \quad (49)$$

Finally, using

$$V_Q = \sum_{j=1}^d \mathbb{E}[\|\xi_j\|_2^2], \quad (50)$$

we obtain

$$\text{Var}(\hat{g}_\Delta) = \sum_{j=1}^d \mathbb{E}[\|\xi_j - \tilde{\Delta}_j\|_2^2]. \quad (51)$$

Therefore, the unrestricted minimum is attained at

$$\tilde{\Delta}_j^*(z_j, v) = \xi_j(z_j, v), \quad (52)$$

for each  $j$ .  $\square$